PRIORITY 4

Business Findings & Recommendations

Olist Brazilian E-Commerce Capstone Project

1. Purpose and business question

This document consolidates the key findings from the Olist e-commerce analysis and translates them into practical business recommendations. The central business question is: Are delivery delays affecting customer satisfaction, and what operational improvements should Olist make?

The analysis addresses five supporting questions:

- Does late delivery relate to customer review scores?
- Which customer states experience the highest delivery delays?
- Which product categories and sellers generate the highest merchandise revenue?
- Which payment methods are most widely used, and how does payment value differ?
- Which operational area should management prioritize?

2. Findings by business question

2.1 Delivery timing and customer satisfaction

The analysis included delivered orders with valid actual and estimated delivery dates. Multiple review records were aggregated to an average review score per order, then grouped as Low (1–2), Neutral (3), or High (4–5).

Delivery group	Low (1–2)	Neutral (3)	High (4–5)
Late	4,136 (54.02%)	874 (11.41%)	2,651 (34.62%)
On time	8,100 (9.19%)	7,071 (8.02%)	72,992 (82.79%)

Interpretation: 54.02% of late-delivery orders were classified as low-rated, compared with 9.19% of on-time orders. Conversely, 82.79% of on-time orders were in the high-rating group, compared with 34.62% of late orders. The chi-square test rejects the null hypothesis of independence. The reported software p-value displayed as 0.0 due to numerical underflow; it should be reported as $p < 0.001$, not as an exact zero.

Statistical conclusion: delivery timing and review-rating group are associated in this sample. The result supports prioritizing delivery reliability, while recognizing that other factors (product quality, seller performance, customer service, and expectations) may also influence reviews.

2.2 Geographic delivery performance

The supplied state summary contains 27 states. The highest late-delivery percentages are shown below. These percentages are the rates in the supplied state-level CSV.

State	Delivered orders	Late orders	Average delay (days)	Late delivery rate
AL	397	85	2.04	21.41%
MA	717	125	1.83	17.43%
SE	335	51	2.47	15.22%
PI	476	66	1.85	13.87%
CE	1,279	176	2.09	13.76%
RR	41	5	4.44	12.20%

BA	3,256	396	1.46	12.16%
RJ	12,350	1,495	1.64	12.11%
PA	946	106	1.44	11.21%
ES	1,995	214	1.21	10.73%

Rate and volume tell different stories. AL has the highest rate (21.41%) among the listed states, while SP has the highest absolute late-order count (1,820), followed by RJ (1,495). MG (519) and BA (396) are also significant by late-order volume. RR has the highest average delay (4.44 days), but only 41 delivered orders in the extract, so its small sample should be considered before committing major resources.

Operational implication: use a two-part prioritization approach—target high-rate states for investigation and high-volume states for the largest potential reduction in the number of late orders.

2.3 Product category and seller revenue

Top product categories by merchandise revenue:

Product category	Revenue (R$)	Items	Share of revenue
Health & Beauty	1,255,695.13	9,634	9.31%
Watches & Gifts	1,198,185.21	5,970	8.88%
Bed, Bath & Table	1,035,964.06	11,097	7.68%
Sports & Leisure	979,740.92	8,590	7.26%
Computers & Accessories	904,322.02	7,781	6.70%

The top five categories contribute approximately 39.82% of the merchandise revenue represented in the category summary. Total category revenue in this file is R$13,494,400.74.

Top sellers by merchandise revenue (seller IDs are anonymized identifiers):

Rank	Seller ID (shortened)	Revenue (R$)	Items	Revenue share
1	4869f7a5...f3b52b2	229,237.63	1,155	1.70%
2	53243585...53d8905	222,776.05	410	1.65%
3	4a3ca931...1493884	200,326.12	1,985	1.48%
4	fa1c13f2...2fecda94	192,842.13	585	1.43%
5	7c67e144...6a010ab	187,923.89	1,364	1.39%

The top 10 sellers together account for approximately 13.20% of merchandise revenue (R$1,781,770.06). Revenue is distributed across many sellers rather than being concentrated in only a few. Seller IDs are technical identifiers and should not be presented as seller names.

2.4 Payment method usage and value

Payment method	Payment records	Payment value (R$)	Average per record (R$)	Share of value
Credit card	76,795	12,542,084.19	163.32	78.34%
Boleto	19,784	2,869,361.27	145.03	17.92%
Voucher	5,775	379,436.87	65.70	2.37%
Debit card	1,529	217,989.79	142.57	1.36%
Not defined	3	0.00	0.00	0.00%

Credit cards dominate the supplied payment extract, representing 78.34% of total payment value and 76,795 payment records. Boleto is the second-largest method by value share at 17.92%. Voucher usage is smaller, and its average payment amount is the lowest among the main payment methods.

Important definition: the CSV contains payment records, not necessarily unique orders. An order can have multiple payment records. Therefore, the average amounts above should be described as average payment-record amounts

unless the underlying query explicitly aggregates to one value per order. Do not label them average order value without validating the calculation.

2.5 Management priority

The strongest management priority is improving delivery reliability and reducing late deliveries. This recommendation is supported by the large difference in low-rating proportions between late and on-time orders, together with the statistically significant association. Geographic targeting should combine late-delivery rate and late-order volume. Revenue-category and payment findings should inform commercial protection and customer-experience initiatives, but they do not outweigh the delivery-satisfaction signal as the central operational priority.

3. Business recommendations

1. Improve delivery reliability

Establish a delivery-performance improvement programme focused on reducing late orders. Monitor promised versus actual delivery dates, identify recurring delay causes, and review carrier and fulfilment performance weekly. Set measurable targets for reducing the late-delivery rate and increasing on-time delivery.

2. Prioritize states using both rate and volume

Start root-cause reviews in AL, MA, and SE because they have the highest late-delivery rates in the supplied state summary. In parallel, prioritize SP and RJ for high-volume interventions because they account for the largest absolute numbers of late orders. Treat RR as a diagnostic case rather than an immediate large-scale investment because its order count is small.

3. Strengthen seller and carrier performance management

Create seller/carrier scorecards covering on-time delivery rate, average delay, order volume, cancellation rate, and customer review outcomes. Work with high-volume or persistently underperforming partners on dispatch speed, handover timing, packaging readiness, and carrier selection. Use seller-level comparisons to investigate performance rather than assuming high revenue means poor or good delivery performance.

4. Improve customer communication when delays occur

Send proactive, accurate delivery-status updates when an order is at risk of arriving late. Provide revised estimated delivery dates, explain the next step, and make support easy to reach. Track whether proactive notifications reduce complaints, support contacts, and low review scores.

5. Protect high-revenue product categories

Maintain availability and fulfilment capacity for Health & Beauty, Watches & Gifts, Bed, Bath & Table, Sports & Leisure, and Computers & Accessories. These five categories contribute about 39.82% of the category revenue in the supplied extract. Review stock readiness, seller coverage, packaging requirements, and delivery promises for these categories.

6. Optimize the payment experience

Keep the credit-card checkout journey reliable because credit cards represent 78.34% of payment value. Maintain boleto as an important alternative, monitor payment failures and abandonment by method, and review the voucher journey given its lower average payment-record amount. Evaluate payment performance using both record-level and order-level metrics.

4. Final conclusion

The analysis indicates that delivery reliability is the most important operational area for Olist to address. Late-delivered orders show a substantially higher proportion of low customer ratings than on-time orders, and the relationship is statistically significant. Olist should improve delivery execution, target geographic areas using both late-delivery rates and absolute late-order volumes, strengthen seller/carrier performance management, and communicate proactively with customers when delays occur. At the same time, it should protect fulfilment for leading revenue categories and maintain a smooth credit-card and boleto payment experience. Before final publication, reconcile the state-level late-order totals and the payment-value discrepancy between the CSV analysis and Power BI.